01

PROBLEM STATEMENT

& CONTEXT

1.5 Billion

civil disputes unresolved annually

4 Years

avg. Indian court case

Rs.50K+

min. legal fees per dispute

Only 11%

of disputes ever reach formal resolution

Justice shouldn't require a lawyer or a fortune.

Who faces this problem?

Employees

Face unpaid wages and wrongful terminations but cannot afford legal representation — most silently absorb the loss.

Tenants

Landlords routinely withhold security deposits and conduct illegal evictions, exploiting the power asymmetry.

Consumers

Victims of fraud and defective products face a slow, bureaucratic redressal system with uncertain outcomes.

Small Businesses

Contract breaches are common, yet litigation costs frequently exceed the disputed amount — making court pointless.

The Real Gap

Lawyers are unaffordable. Courts take 4-6 years. Existing apps lack true AI — they provide forms, not resolution.

02

SOLUTION OVERVIEW

LegalAI Resolver

What Is LegalAI Resolver?

LegalAI Resolver is a fully automated pre-litigation platform that converts any plain-English dispute description into a complete, legally-structured resolution — without lawyers, court filings, or delays.

The platform orchestrates 5 specialized AI agents in a sequential pipeline:

- Intake Agent: classifies dispute type, extracts facts, detects severity
- Legal Agent: maps applicable statutes with jurisdiction and section numbers
- Analytics Agent: models win probability, costs, and settlement range
- Document Agent: generates 3 separate legal PDFs (Demand Letter, Court File, Settlement Agreement)
- Negotiation Agent: mediates a fair settlement across up to 3 rounds

End result: a signed settlement agreement delivered to both parties in minutes — not months.

SPEED

Minutes

vs. 6 – 18 months through court proceedings

COST

Near Zero

vs. Rs.50,000+ in legal representation fees

AI INTELLIGENCE

5 Agents

Intake + Law research + analytics + mediation + documents

"Describe your dispute in plain English. Get applicable law, outcome probability, and a signed settlement — all automated."

03

KEY FEATURES & USER FLOW

1. Describe Dispute
2. AI Intake & Parse
3. Legal Research
4. Analytics & Odds
5. AI-Guided Mediation
6. Settlement Agreement

Multi-Agent AI Pipeline

5 specialized agents run sequentially — each receives the prior agent's output as context, creating an intelligent cascading analysis chain.
Agents : Intake > Legal Research > Analytics > Document > Negotiation

Automated Legal Research

Maps applicable laws with section numbers, jurisdiction, relevance scores, and claimant rights. Covers Indian (IPC, TPA, CPA), US, and UK statutes across 12 dispute types.

Probabilistic Case Analytics

Win probability (0-100%), court filing fee estimate, legal cost projection, expected timeline, confidence level, and evidence-based recommended settlement range.

AI Mediator — 3-Round System

Claimant and respondent submit offers independently each round. AI Mediator proposes a fair amount and reasoning. Settlement finalizes only when BOTH parties explicitly Accept.

3 Separate Legal Documents

Each with its own dedicated AI agent and visual identity: Navy — Demand Letter (5-paragraph legal notice). Red — Court-Ready File (evidence checklist, filing steps, witnesses). Green — Settlement Agreement (9 clauses, dual signatures).

Azure-Powered Email System

Respondent receives branded invite with unique secure portal link. AI proposal emails sent each round. Settlement agreement delivered with PDF download link to both parties.

04 ARCHITECTURE & TECHNOLOGY APPROACH

FRONTEND

HTML · Tailwind CSS · Vanilla JS

- index.html — Claimant portal: dispute submission, 4-tab results dashboard
- respond.html — Respondent portal: offer submission, accept/reject proposals
- Deployed on Render

HTTP / REST

BACKEND (FastAPI + Python 3.11)

- 14 REST endpoints: /analyze /negotiate /proposal-response /my-cases
- SQLite + db_service.py — full case & session persistence
- Azure Communication Services — transactional email at every stage
- ReportLab — 3 separate PDF generators (Demand, Court, Settlement)
- ngrok — secure public HTTPS tunnel for live demo

AI AGENT STACK

1 Intake Agent

Classifies 12 dispute types · Extracts key facts · Detects severity (LOW/MEDIUM/HIGH/CRIMINAL) · Separates monetary vs. non-monetary · Structures full JSON case object

2 Legal Agent

Maps applicable laws with section numbers, jurisdiction, relevance scores, and claimant rights · Identifies respondent defenses · India, US, UK coverage

3 Analytics Agent

Win probability 0–100% · Court filing fee estimate · Legal cost projection · Confidence level · Recommended settlement range (min/optimal/max)

4 Document Agent

3 dedicated sub-agents: Demand Letter (5-paragraph legal notice) · Court-Ready File (facts, evidence checklist, filing steps, witnesses) · Settlement Agreement (9 clauses)

5 Negotiation Agent

Analyzes both parties' offers each round · Proposes a fair amount with full reasoning · Gap analysis · Pressure points for both sides · Max 3 rounds

05 DESIGN DECISIONS & TRADE-OFFS

DECISIONS MADE

Sequential Agent Design

Each agent receives the prior agent's full output as context, creating an intelligent cascading chain of analysis.

Unicode-Safe 3-Agent PDF System

3 separate AI agents with distinct prompts + PDF layouts. Custom _safe() function pre-converts Rs., EUR etc. — fixes silent ReportLab crash on all currency symbols

3-Round Negotiation Cap

Mirrors professional mediation practice, forces convergence, and prevents indefinite back-and-forth cycling.

SQLite over PostgreSQL

Zero-config, file-based storage is sufficient for prototype scale and eliminates infrastructure complexity.

INTENTIONALLY DROPPED

User Authentication

Email-based case linking was sufficient for MVP scope; full auth adds friction without meaningful value at this stage.

Multi-language Support

Prioritised English for MVP clarity and consistency; localisation is a clean post-hackathon addition.

Human Lawyer Review Loop

AI-only pipeline keeps cost near zero — adding mandatory human review negates the core value proposition.

Native Mobile App

A responsive web interface addresses 90% of use cases; a native app is premature for a prototype submission.

CONSTRAINTS FACED

LLM Accuracy Limits

JSON-mode prompting and low temperature reduce errors; however, legal accuracy is not formally verified.

No Legal Enforceability

Settlement agreements are AI-generated drafts and are not substitutes for lawyer-reviewed legal documents.

API Rate Limits

Groq's free tier has per-minute request caps — production deployment would require a paid tier upgrade.

06 CURRENT STATUS, LIMITATIONS & NEXT STEPS

FULLY FUNCTIONAL

- Full dispute-to-settlement end-to-end flow
- All 5 AI agents with structured JSON output
- 3-round negotiation with Accept / Reject per party
- 3 separate PDFs with 3 dedicated AI agents
- Unicode-safe PDF generation (Rs. fix applied)
- Respondent email invites via Azure Comm. Services
- Settlement PDF auto-generated and emailed
- Secure respondent portal (respond.html)
- Non-monetary disputes (apology, harassment, assault)
- Analysis-only mode — no respondent needed
- Email-based My Cases history dashboard
- Criminal severity detection + police advisory

KNOWN LIMITATIONS

- Legal accuracy not formally verified by attorneys
- Settlement agreements are AI-drafted — not lawyer-reviewed
- PDFs lost on server restart (ephemeral storage)
- SQLite not suitable for concurrent multi-user production
- No input sanitization or rate limiting yet
- No email verification before case submission
- Mobile layout partially tested on respond.html

PLANNED NEXT STEPS

- Lawyer review layer for settlement agreements
- WhatsApp / SMS notifications via Azure
- 10+ Indian language support
- B2B API for legal-tech platform integrations
- Blockchain-anchored settlement records
- Digital signatures via Azure or DocuSign
- Automated court filing document preparation